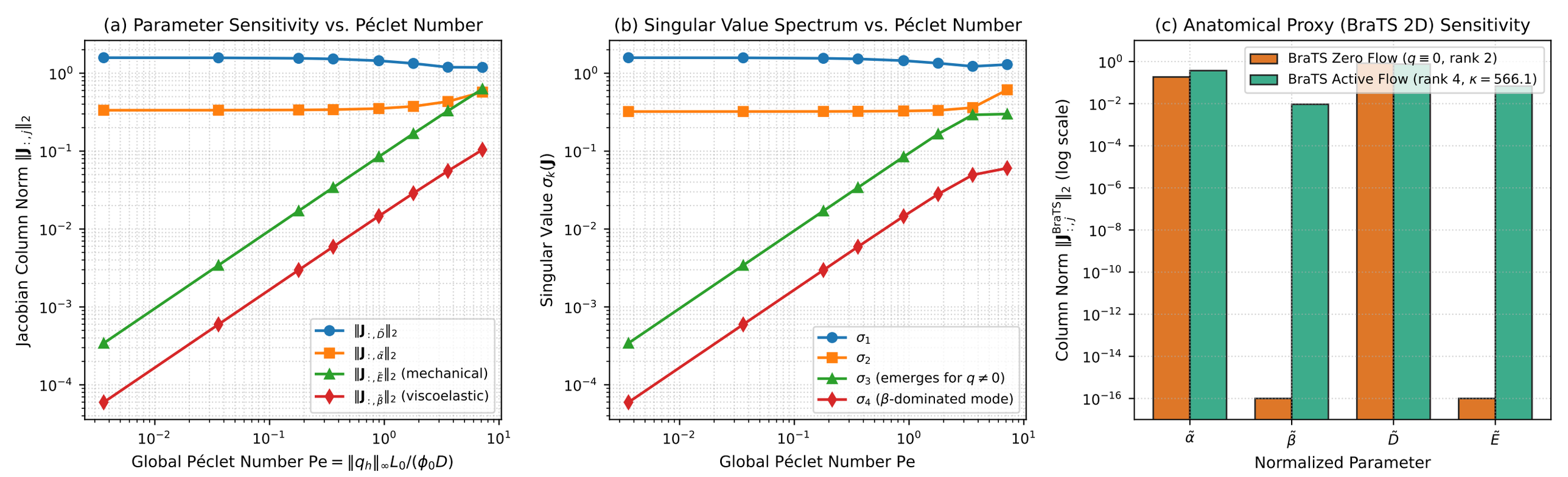
(a) Parameter Sensitivity vs. Péclet Number
Jacobian Column Norm $\|\mathbf{J}_{:,j}\|_2$
Global Péclet Number Pe $= \|q_h\|_\infty L_0/(\phi_0 D)$
$\|\mathbf{J}_{:,\tilde{D}}\|_2$
$\|\mathbf{J}_{:,\tilde{\alpha}}\|_2$
$\|\mathbf{J}_{:,\tilde{E}}\|_2$ (mechanical)
$\|\mathbf{J}_{:,\tilde{\beta}}\|_2$ (viscoelastic)
(b) Singular Value Spectrum vs. Péclet Number
Singular Value $\sigma_k(\mathbf{J})$
Global Péclet Number Pe
σ_1
σ_2
σ_3 (emerges for $q \neq 0$)
σ_4 (β-dominated mode)
(c) Anatomical Proxy (BraTS 2D) Sensitivity
Column Norm $\|\mathbf{J}^{\mathrm{BraTS}}_{:,j}\|_2$ (log scale)
BraTS Zero Flow ($q \equiv 0$, rank 2)
BraTS Active Flow (rank 4, $\kappa = 566.1$)
$\tilde{\alpha}$
$\tilde{\beta}$
$\tilde{D}$
$\tilde{E}$
Normalized Parameter